

SEA

É. Savin

What is it
all about?

Course
description

Bibliography

Advanced structural dynamics and acoustics

Introduction à l'Analyse Statistique Énergétique (SEA)

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Outline

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1 What is it all about?

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3 Bibliography

Random loads

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Launcher structure
excited by jet noise



Aircraft structure excited by
turbulent boundary layer

Random loads

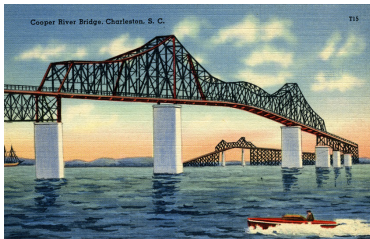
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Bridge excited by wind



Offshore oil rig
excited by swell

Random loads

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Earthquakes



Tank excited by
road unevenness

Frequency ranges of vibration

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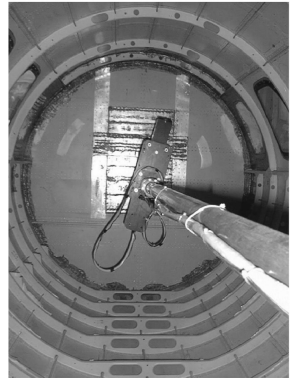
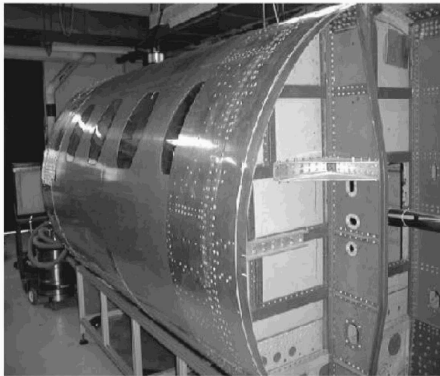
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- Cessna Citation fuselage: 9+19 stringers, 22 stiffeners.
- $L = 2.55$ m, $R = 0.81$ m, thickness = 0.8–1.2 mm.



Herdic *et al.* *JASA* 117(6), 3667 (2005)

Frequency ranges of vibration

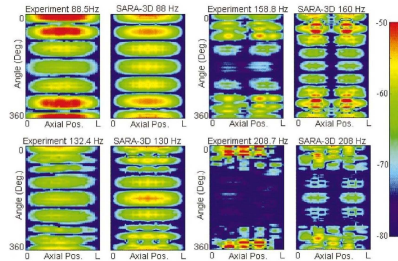
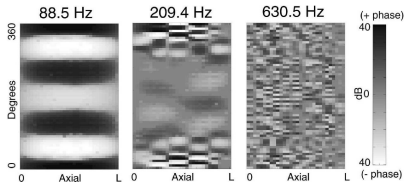
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Phase of the radial celerity

Ampl. of the radial celerity

Loading: a point force at the junction of a frame and stiffener.

Frequency ranges of vibration

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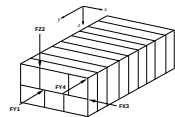
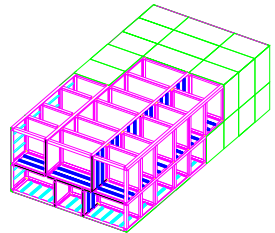
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- Structure: 200 plates, 400 stiffeners, 54 cavities.
- $L = 5.3$ m, width = 2.5 m, height = 1.4 m, $M = 825$ kg.



Savin *AIAA J.* **40**(9), 1876 (2002)

Frequency ranges of vibration

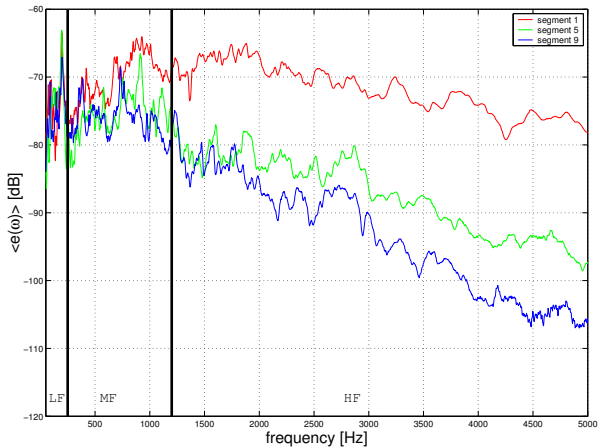
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$$\text{Estimated mechanical energies } \mathcal{E} = m\langle v \rangle^2;$$
$$dB_{\text{ref}} = 10 \times \log_{10}(1 \text{ kg} \cdot \text{m}^2/\text{s}^2).$$

Frequency ranges of vibration

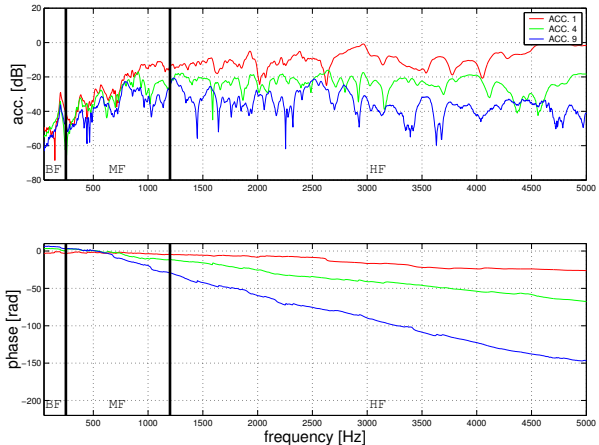
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Measured vertical accelerations along the structure;
$$dB_{\text{ref}} = 20 \times \log_{10}(1 \text{ m/s}^2).$$

Frequency ranges of vibration

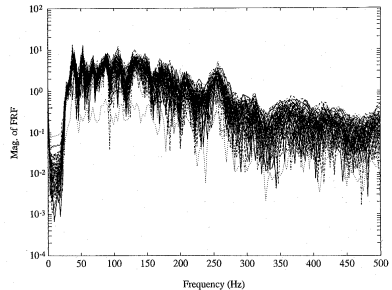
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Measured FRF amplitudes of 99 "identical" Isuzu Rodeo trucks: acoustic pressure at driver's ear for a mechanical excitation applied to the left front wheel.

Kompella-Bernhard *Noise Control Engrg. J.* 44(2), 93 (1996)

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- At **low frequencies** the dynamic response of a built-up structure is physically well represented by component mode synthesis (Structural dynamics and vibration courses).
- At **higher frequencies** this representation is mathematically sound, but numerically and above all **physically** meaningless. These issues have critical relevance in both internal and external structural acoustics (Fluid-structure interaction, Structural acoustics, Boundary element method courses).

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- SEA is to date the only one engineering approach capable of dealing with them, but it is based on (very) restrictive hypotheses. The objectives of the course are twofold:
 - 1 introduce the basic SEA concepts,
 - 2 discuss the underlying assumptions and limitations.
- Use of a **probabilistic framework** (Statistical learning course).

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- **Prerequisites:** structural dynamics, structural acoustics, fluid-structure interaction, random vibrations, elastic wave propagation:
 - Vibrations, analyse modale (É. Balmès, M. Corus);
 - Méthodes fréquentielles en vibrations (A. Barbarulo);
 - Interaction fluides-structures (J.-S. Schotté);
 - Propagation des ondes élastiques (B. Tie);
 - Apprentissage statistique (D. Clouteau, F. Gatti).

but all lectures are self-contained!!!

- **Lecture notes & slides:** on demand
`eric.savin@centralesupelec.fr`.
- **Edunao:**
<https://centralesupelec.edunao.com/course/view.php?id=10825>
- **Grading:** final exam, homework.

Agenda 2025–2026

All lectures start at 09:00 AM except #7, #8 at 13:45 PM

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11 Sep **Lecture #1:** primer on [random vibrations](#) & energetics of the single-dof oscillator.

25 Sep **Lecture #2:** fundamental coupled problems:

1 2-dofs system [after R.H. Lyon, G. Maidanik, *JASA* **34**(5), 623 (1962)],

2 single-dof oscillator in an acoustic fluid [after P.W. Smith Jr., *JASA* **34**(5), 640 (1962)].

02 Oct **Lecture #3:** [tutorial class #1](#)

09 Oct **Lecture #4:** energetics of [continuous system vibrations](#).

16 Oct **Lecture #5:** energetics of [coupled system vibrations](#); structural-acoustics bases, inertial and radiation effects.

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23 Oct **Lecture #6:** tutorial class #2.

06 Nov **Lecture #7:** structural-acoustics (cont'd): infinite
and finite plates coupled to the air.

13 Nov **Lecture #8:** SEA formulation, hypotheses,
shortcomings, parameters. Industrial applications.

20 Nov **Lecture #9:** tutorial class #3.

11 Dec **Lecture #10:** external Neumann problem for the
Helmholtz equation.

TBA **Final exam.**

Other graduate courses

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- **M.I.T. 13.811:** Advanced structural dynamics and acoustics [available on [MIT OCW](#)].
- **U. of Adelaide MECH ENG 7030:** Advanced vibrations.
- **K.T.H. SD2170:** Energy methods.
- **U. of Illinois TAM 514 ONL:** Elastodynamics and Vibrations.
- **Boston U. ENG ME 515:** Vibrations of complex mechanical systems.
- **Shandong U. Postgraduate:** Mechanical vibration.
- *etc.*

History

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- M. Heckl 1930–1996, R.H. Lyon 1929–2019, **G. Maidanik 1925–2010**, D.U. Noiseux 1920–2009, P.W. Smith, Jr. 1923–2010, E.E. Ungar 1926–, **W. Westphal 1928–???...**
- Bolt, Beranek & Newman Inc., Cambridge Acoustical Associates Inc., Harvard Acoustical Research Lab., MIT Acoustics and Vibration Lab. (Cambridge MA), Naval Ship Research and Development Center (Bethesda MD)...
- Apollo program 1961–1975, acoustic signature of ships and submarines, aeroacoustics, underwater acoustics...

Further reading...

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- J. S. Bendat, A. G. Piersol: *Random Data: Analysis and Measurement Procedures*, 4th Ed. Wiley, Hoboken NJ (2010).
- C. F. Coe, W. J. Chyu: "Pressure-fluctuation inputs and response of panels underlying attached and separated supersonic turbulent boundary layers", *AGARD Symposium of Acoustic Fatigue*, 26-27 September 1972, Toulouse; 20 pages (1972).
- G. M. Corcos: "Resolution of pressure in turbulence", *J. Acoust. Soc. Am.* **35**(2), 192-199 (1963).
- F. Fahy, P. Gardonio: *Sound and Structural Vibration: Radiation, Transmission and Response*, 2nd Ed. Academic Press, London (2006).
- M.C. Junger, D. Feit: *Sound, Structures, and their Interaction*, 2nd Ed. Acoustical Society of America, Melville NY (1993).
- H. Kobayashi, B. L. Mark, W. Turin: *Probability, Random Processes, and Statistical Analysis*. Cambridge University Press, Cambridge (2012).

Further reading...

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- A. Le Bot: *Foundation of Statistical Energy Analysis in Vibroacoustics*. Oxford University Press, Oxford (2015).
- R.H. Lyon, R.G. DeJong: *Theory and Applications of Statistical Energy Analysis*, 2nd Ed. Butterworth-Heinemann, Boston MA (1995).
- G. Maidanik: "Vibrational and radiative classifications of modes of a baffled finite panel", *J. Sound Vib.* **34**(4), 447-455 (1974).
- R. Ohayon, C. Soize: *Structural Acoustics and Vibration*. Academic Press, London (1998).
- J. J. Shynk: *Probability, Random Variables and Random Processes. Theory and Signal Processing Applications*. Wiley, Hoboken NJ (2013).
- P.W. Smith Jr., R.H. Lyon: *Sound and Structural Vibration*. NASA CR-160, March 1965.